⭐⭐⭐⭐⭐ 1. Read, decode and make sense of statements... (Item 4)

Most critical

- Students must correctly understand that:
  - 100 mL is the **total volume**
  - The ingredient amounts sum to 100 mL
  - The task is to **keep the ratio unchanged**, not to create a new recipe
- This is the **first step of mathematical modelling from a real-world context**
- Without correct interpretation of the text, no mathematical modelling can occur

👉 This reflects **constructing a mathematical model from a real-world situation**

⭐⭐⭐⭐⭐ 2. Recognise mathematical structure (Item 5)

Very important

- Students need to recognise that:
  - The ingredients are in a **fixed proportional relationship**
  - Increasing the total volume from 100 mL to 150 mL represents a **uniform scaling**
- The implicit structure is:
  - Salad oil makes up **60%** of the total
  - Or the **scale factor = 150 ÷ 100 = 1.5**

👉 This demonstrates recognition of a **proportional structure / direct variation**

⭐⭐⭐⭐ 3. Recognise aspects corresponding with known mathematical concepts (Item 8)

Highly relevant

- The problem clearly corresponds to familiar concepts:
  - Ratios
  - Percentages
  - Enlargement and scaling
- Students are expected to realise:
  - "This is a standard proportional reasoning problem"

👉 This shows the ability to **connect a new context with known mathematical ideas**

⭐⭐⭐⭐ 4. Identify and describe the mathematical aspects of a real-world situation (Item 6)

Important

- Key mathematical elements include:
  - Total volume (100 mL, 150 mL)
  - Amount of salad oil
  - The invariant ratio
- Students must distinguish:
  - Which quantities are **mathematically significant**
  - Which information is contextual background (e.g. "salad dressing")

👉 This reflects identifying **mathematical features embedded in a real context**

⭐⭐⭐⭐ 5. Identify the key variables in a model (Item 2)

Important

Key variables include:

- Total volume of dressing
- Amount of salad oil
- Scaling factor / ratio

👉 This forms a **simple proportional model**

⭐⭐⭐ 6. Translate a problem into a standard mathematical representation or algorithm (Item 9)

Moderately high relevance

- The problem can be translated into:
  - $60 \div 100 \times 150$
  - or 60×1.5
- This conversion from words to an equation is a **typical modelling step**

⭐⭐⭐ 7. Simplify or decompose a situation (Item 7)

Moderate relevance

- Students can ignore unnecessary details:
  - Vinegar and soy sauce do not need to be calculated
- The situation is simplified into:
  - A single proportional calculation for salad oil

👉 This is an example of **reducing a real situation to its essential mathematical core**

⭐⭐⭐ 8. Identify constraints, assumptions, and simplifications in a model (Item 12)

Moderate relevance

Implicit assumptions include:

- The recipe ratio remains **exactly the same**
- There is no waste or measurement error
- Ingredient quantities change linearly

👉 These are typical **modelling assumptions**

⭐⭐ 9. Select a mathematical description or representation (Item 1)

Lower relevance

- Possible representations include:
  - Percentages
  - Ratios
  - Scale factors
- However, the problem does not strongly require choosing among multiple representations

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⭐⭐ 10. Use mathematical tools to describe structures (Items 10 & 11)

Low relevance

- Only basic numerical calculations are needed
- No diagrams, symbols, or advanced representations are required

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⭐ 11. Select a representation appropriate to the problem context (Item 3)

Lowest relevance

- The proportional representation is essentially predetermined by the problem
- Students have limited choice in representation

⭐⭐⭐⭐⭐ 1. Read, decode and make sense of statements…（第4条）

最核心

- 学生需要正确理解：
  - 100 mL 是**总量**
  - 各成分之和等于 100 mL
  - 问题不是重新配方，而是**保持配比不变**
- 这是**从文字情境建模的第一步**
- 如果不能正确解读文本，就无法进入数学建模

👉 体现了**从现实语境构建数学情境模型**

⭐⭐⭐⭐⭐ 2. Recognise mathematical structure（第5条）

非常关键

- 学生需要识别：
  - 各成分之间是**固定比例关系**
  - 总量从 100 → 150，是**整体等比例放大**
- 隐含结构是：
  - 沙拉油占总量的 **60%**
  - 或 “放大倍数 = 150 ÷ 100 = 1.5”

👉 这是对**比例结构 / 正比例关系**的识别

⭐⭐⭐⭐ 3. Recognise aspects corresponding with known concepts（第8条）

高度相关

- 该问题可直接对应到熟悉的数学知识：
  - 比例
  - 百分比
  - 放大缩小
- 学生会意识到：
  - “这是一个典型的按比例计算问题”

👉 属于**把新情境连接到已知数学概念**

⭐⭐⭐⭐ 4. Identify and describe the mathematical aspects…（第6条）

重要

- 数学要素包括：
  - 总量（100 mL、150 mL）
  - 成分用量（沙拉油）
  - 配比不变
- 学生需区分：
  - 哪些是**重要变量**
  - 哪些信息只是背景（如“沙拉酱”）

👉 明确了**现实问题中的数学成分**

⭐⭐⭐⭐ 5. Identify the key variables in a model（第2条）

重要

关键变量包括：

- 总量（100 mL → 150 mL）
- 沙拉油用量
- 放大倍数 / 比例

👉 建立了一个**简单比例模型**

⭐⭐⭐ 6. Translate a problem into a standard mathematical representation（第9条）

中等偏高相关

- 学生可以将问题转化为：
  - $60 \div 100 \times 150$
  - 或 60×1.5
- 从文字 → 数学算式，是典型的建模翻译过程

⭐⭐⭐ 7. Simplify or decompose the situation（第7条）

中等相关

- 实际上学生会忽略不必要信息：
  - 不需要分别计算醋和酱油
  - 只关注"沙拉油占比"
- 把复杂情境**简化为一个比例问题**

⭐⭐⭐ 8. Identify constraints, assumptions, simplifications（第12条）

中等相关

隐含假设包括：

- 配方比例**保持完全不变**
- 不存在浪费或测量误差
- 各成分线性变化

👉 学生默认接受这些建模前提

⭐⭐ 9. Select a mathematical description or representation（第1条）

一般相关

- 可以选择：
  - 百分比表示
  - 比例表示
  - 放大倍数
- 但题目本身并未强制要求学生在多种表示间选择

⭐⭐ 10. Use mathematical tools to describe structures（第10 / 11条）

较低相关

- 只涉及非常基础的算式
- 不需要图表、变量符号或复杂表示

⭐ 11. Select a representation appropriate to the problem context（第3条）

相关性最低

- 题目已隐含“比例”是最自然的表示
- 学生选择空间不大